

Structure of Detailed Design Document

1. Introduction

Give description of the Detailed Design

2. Applicable Documents

The documents used during detailed design are:

- ☐ System Requirements Document
- ☐ System Design
- ☐ Database Design

The detailed design refines the system document. Hence the first applicable document here is system design. Also we are referring the data structure. Hence the second applicable document here is database design.

3. Structure of the software package

The functional components identified during the system design are listed here.

- ☐ Functional component 1
- ☐ Functional component 2:
- ☐ Functional component 3:
- ☐ Functional component 4:
- ...

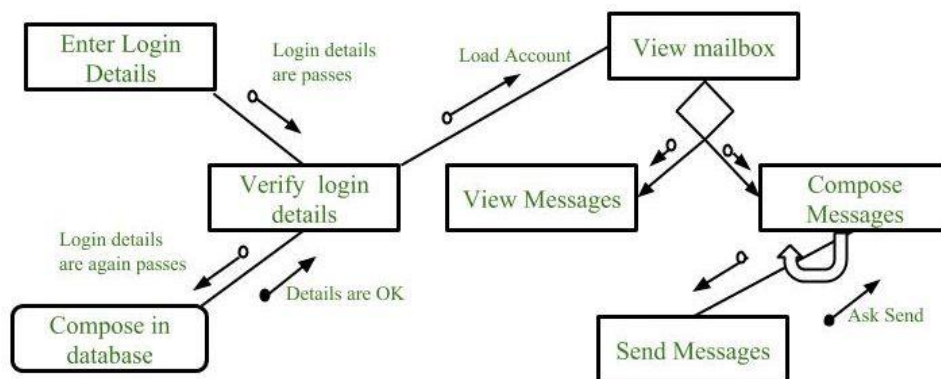
4. Modular Decomposition of components

4.1. Modules of component 1

4.1.1. Design assumptions, if any

4.1.2. Identification of modules

4.1.3. Structure chart showing the hierarchy of modules (calling and called modules)



<https://www.geeksforgeeks.org/software-engineering-structure-charts/>

4.1.4. Data structures shared among modules

4.2. Modules of component 2

4.2.1. Design assumptions, if any

4.2.2. Identification of modules

4.2.3. Structure chart showing the hierarchy of modules (calling and called modules)

4.2.4. Data structures shared among modules

...

4.n Modules of component n

...

5. Detailed Design

5.1. Module Design of component 1

5.1.1. Module 1

5.1.1.1. Inputs (source, format, quantity, units, timing, valid range)

5.1.1.2. Procedural details (Flow Chart / Pseudo-code in Structured English)

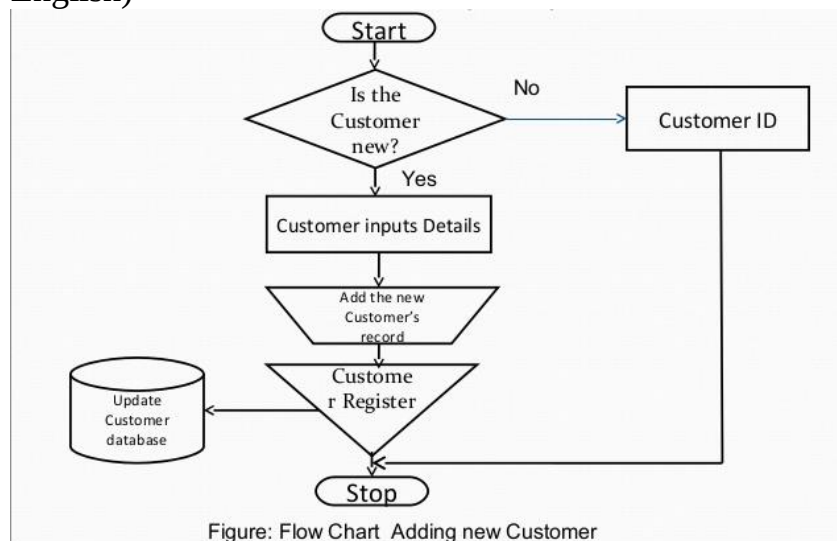


Figure: Flow Chart Adding new Customer

Add Two Numbers Program Pseudocode Algorithm

Declare **Number1, Number2, Sum** As Variables

When the *flag* is clicked

Initialize all variables to 0

Output: "Enter the first number"

Set **Number1** = user answer

Ask user: "Enter the second number:"

Set **Number2** = user answer

Set **Sum** = **Number1** + **Number2**

Output: "The sum is:"

Output: **Sum**

- 5.1.1.3. Table I/O interfaces
- 5.1.1.4. Outputs (destination, format, quantity, units, timing, valid range)
- 5.1.1.5. Implementation aspects, if any (using existing library etc)
- 5.1.2. Module 2
 - 5.1.2.1. Inputs (source, format, quantity, units, timing, valid range)
 - 5.1.2.2. Procedural details (Flow Chart / Pseudo-code in Structured English)
 - 5.1.2.3. File I/O interfaces
 - 5.1.2.4. Outputs (destination, format, quantity, units, timing, valid range)
 - 5.1.2.5. Implementation aspects, if any (using existing library etc)
 - ...
- 5.2. Module Design of component 2
 - 5.2.1. Module 1
 - 5.2.1.1. Inputs (source, format, quantity, units, timing, valid range)
 - 5.2.1.2. Procedural details (Flow Chart / Pseudo-code in Structured English)
 - 5.2.1.3. File I/O interfaces
 - 5.2.1.4. Outputs (destination, format, quantity, units, timing, valid range)
 - 5.2.1.5. Implementation aspects, if any (using existing library etc)
 - 5.2.2. Module 2
 - 5.2.2.1. Inputs (source, format, quantity, units, timing, valid range)
 - 5.2.2.2. Procedural details (Flow Chart / Pseudo-code in Structured English)
 - 5.2.2.3. File I/O interfaces
 - 5.2.2.4. Outputs (destination, format, quantity, units, timing, valid range)
 - 5.2.2.5. Implementation aspects, if any (using existing library etc)
 - ...
- 5.n Module Design of component **n**